

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403841
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Rodriguez; William B.

All-terrain vehicle

Abstract

An all-terrain vehicle includes a frame, a plurality of ground-engaging members for supporting the frame, an outer body supported by the frame, and a cargo rack pivotably coupled to the outer body. The cargo rack is moveably between a closed position and an opened position and includes a hinge assembly comprising hinge leaves and a removable hinge pin.

Inventors:	Rodriguez; William B. (Roseau, MN)
Applicant:	Polaris Industries Inc. (Medina, MN)
Family ID:	76383649
Assignee:	Polaris Industries Inc. (Medina, MN)
Appl. No.:	18/211951
Filed:	June 20, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230331169 A1	Oct. 19, 2023

Related U.S. Application Data

division parent-doc US 16723182 20191220 US 11718240 child-doc US 18211951

Publication Classification

Int. Cl.: B60R9/06 (20060101); B60K1/04 (20190101)

U.S. Cl.:

CPC **B60R9/06** (20130101); **B60K1/04** (20130101); B60K2001/0411 (20130101);
B60Y2200/124 (20130101)

Field of Classification Search

CPC: B60R (9/06); B60K (1/04); B60K (2001/0411); B60Y (2200/124)

USPC: 180/233

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1082423	12/1912	Hartman	N/A	N/A
1951220	12/1933	Tarr	N/A	N/A
2141093	12/1937	Edward et al.	N/A	N/A
2284352	12/1941	Zank	N/A	N/A
2396511	12/1945	Alexander	N/A	N/A
2628657	12/1952	Orrick, Jr.	N/A	N/A
2672103	12/1953	Hohmes	N/A	N/A
2868273	12/1958	Barrett	N/A	N/A
2984290	12/1960	Miller	N/A	N/A
3167298	12/1964	Alexander et al.	N/A	N/A
3347512	12/1966	Campbell	N/A	N/A
3508764	12/1969	Dobson et al.	N/A	N/A
3560022	12/1970	Gold	N/A	N/A
3736020	12/1972	Pilachowski et al.	N/A	N/A
3747888	12/1972	Heckett	N/A	N/A
3841696	12/1973	Wagner	N/A	N/A
3844610	12/1973	Adams	N/A	N/A
3858902	12/1974	Howells et al.	N/A	N/A
3913975	12/1974	Carter	N/A	N/A
3927854	12/1974	Carey	N/A	N/A
4027892	12/1976	Parks	N/A	N/A
4059171	12/1976	Pakosh	N/A	N/A
4062585	12/1976	Herring, Jr.	N/A	N/A
4098414	12/1977	Abiera	N/A	N/A
4129198	12/1977	Hunter	N/A	N/A
4198092	12/1979	Federspiel	N/A	N/A
4380352	12/1982	Diffrient	N/A	N/A
4521053	12/1984	De Boer	N/A	N/A
4522454	12/1984	Hochstatter	N/A	N/A
4527831	12/1984	Katsuoka	N/A	N/A
4561323	12/1984	Stromberg	N/A	N/A
4563038	12/1985	Hirose	N/A	N/A
4662597	12/1986	Jecker et al.	N/A	N/A
4714227	12/1986	Holm et al.	N/A	N/A
4722031	12/1987	Matsuyama	74/551.8	B62K 5/027
4779695	12/1987	Yasui	N/A	N/A

4821825	12/1988	Somerton-Rayner	N/A	N/A
4844309	12/1988	Aubin et al.	N/A	N/A
4934737	12/1989	Nakatsuka	N/A	N/A
4941784	12/1989	Flament	N/A	N/A
5007675	12/1990	Musto et al.	N/A	N/A
5036939	12/1990	Johnson	192/38	F16D 41/105
5037155	12/1990	Holm et al.	N/A	N/A
5083632	12/1991	Saito et al.	N/A	N/A
5149034	12/1991	Ganaja	N/A	N/A
5251713	12/1992	Enokimoto	N/A	N/A
5309861	12/1993	Mardikian	N/A	N/A
5324095	12/1993	Yamauchi	N/A	N/A
5367978	12/1993	Mardikian	N/A	N/A
5455751	12/1994	Lynch	D26/44	F21V 21/06
5458213	12/1994	Nakaya et al.	N/A	N/A
5509496	12/1995	Erickson et al.	N/A	N/A
5549428	12/1995	Yeatts	N/A	N/A
5613570	12/1996	Becker	N/A	N/A
5618021	12/1996	Brodersen	N/A	N/A
5676292	12/1996	Miller	N/A	N/A
5713629	12/1997	Plackis	N/A	N/A
5776568	12/1997	Andress et al.	N/A	N/A
5863277	12/1998	Melbourne	N/A	N/A
5876085	12/1998	Hill	N/A	N/A
5950750	12/1998	Dong et al.	N/A	N/A
6007150	12/1998	Clerkin et al.	N/A	N/A
6092877	12/1999	Rasidescu et al.	N/A	N/A
6102466	12/1999	Kanazawa et al.	N/A	N/A
6149540	12/1999	Johnson et al.	N/A	N/A
6182590	12/2000	Patera	N/A	N/A
6196634	12/2000	Jurinek	N/A	N/A
6199894	12/2000	Anderson	N/A	N/A
6276653	12/2000	Traxler	N/A	N/A
6354556	12/2001	Ritchie et al.	N/A	N/A
6428192	12/2001	Chen	403/398	B62J 6/029
6460818	12/2001	Garelick et al.	N/A	N/A
6467787	12/2001	Marsh	N/A	N/A
6530730	12/2002	Swensen	N/A	N/A
6565241	12/2002	Glynn	362/183	F21L 4/085
6622968	12/2002	St et al.	N/A	N/A
6648569	12/2002	Douglass et al.	N/A	N/A
6695566	12/2003	Rodriguez Navio	N/A	N/A
6725962	12/2003	Fukuda	N/A	N/A
6732830	12/2003	Gagnon	180/908	B60K 17/22
6752235	12/2003	Bell et al.	N/A	N/A
6767040	12/2003	Freijy	N/A	N/A
6773049	12/2003	Rupiper et al.	N/A	N/A
6776250	12/2003	Kuji	180/908	B62J 9/25
6786526	12/2003	Blalock	N/A	N/A
6799779	12/2003	Shibayama	N/A	N/A

6799781	12/2003	Rasidescu et al.	N/A	N/A
6880483	12/2004	Fedders	N/A	N/A
6916142	12/2004	Hansen et al.	N/A	N/A
6923507	12/2004	Billberg et al.	N/A	N/A
RE38895	12/2004	McLemore	N/A	N/A
6968917	12/2004	Rondeau et al.	N/A	N/A
6971714	12/2004	Hanagan	N/A	N/A
6976720	12/2004	Bequette	N/A	N/A
6978857	12/2004	Korenjak	N/A	N/A
7008015	12/2005	Bischoff	N/A	N/A
7055454	12/2005	Whiting et al.	N/A	N/A
7121371	12/2005	Rondeau et al.	N/A	N/A
7124853	12/2005	Kole, Jr.	N/A	N/A
7125134	12/2005	Hedlund et al.	N/A	N/A
7128341	12/2005	Dahl et al.	N/A	N/A
7137764	12/2005	Johnson	N/A	N/A
7237789	12/2006	Herman	N/A	N/A
7239032	12/2006	Wilson et al.	N/A	N/A
7258192	12/2006	Davis et al.	N/A	N/A
7266431	12/2006	Jackson	701/1	B60Q 1/2657
7331418	12/2007	Audet	N/A	N/A
7377223	12/2007	Toupin et al.	N/A	N/A
7416234	12/2007	Bequette	N/A	N/A
7431024	12/2007	Buchwitz et al.	N/A	N/A
7506714	12/2008	Davis et al.	N/A	N/A
D593003	12/2008	Sanschagrin et al.	N/A	N/A
D593454	12/2008	Sanschagrin et al.	N/A	N/A
D594388	12/2008	Sanschagrin et al.	N/A	N/A
D606900	12/2008	Flores	N/A	N/A
7682115	12/2009	Jay et al.	N/A	N/A
7686368	12/2009	Ghergheli et al.	N/A	N/A
7703826	12/2009	German	N/A	N/A
D625662	12/2009	Li	N/A	N/A
8122993	12/2011	Ripley et al.	N/A	N/A
D657720	12/2011	Eck et al.	N/A	N/A
8156841	12/2011	Yasuda et al.	N/A	N/A
8205910	12/2011	Leonard et al.	N/A	N/A
8215694	12/2011	Smith et al.	N/A	N/A
D665309	12/2011	Lepine et al.	N/A	N/A
D665312	12/2011	Lepine et al.	N/A	N/A
8245657	12/2011	Ross et al.	N/A	N/A
D670198	12/2011	Li et al.	N/A	N/A
8376441	12/2012	Nakamura et al.	N/A	N/A
D679627	12/2012	Li et al.	N/A	N/A
8430442	12/2012	Utke et al.	N/A	N/A
D682737	12/2012	Li et al.	N/A	N/A
D682738	12/2012	Li et al.	N/A	N/A
8555799	12/2012	Otsuka et al.	N/A	N/A
D694668	12/2012	Li et al.	N/A	N/A
8596398	12/2012	Bennett	180/90.6	B62K 5/01

D699627	12/2013	Tang	N/A	N/A
D701143	12/2013	Shan	N/A	N/A
8678464	12/2013	Smith et al.	N/A	N/A
8678976	12/2013	Wilson et al.	N/A	N/A
8720355	12/2013	Aoyama	N/A	N/A
8905435	12/2013	Ripley et al.	N/A	N/A
D722927	12/2014	Schroeder et al.	N/A	N/A
D727807	12/2014	Schroeder et al.	N/A	N/A
9080514	12/2014	Utke et al.	N/A	N/A
D756845	12/2015	Flores	N/A	N/A
9649928	12/2016	Danielson	N/A	B60K 11/06
D805009	12/2016	Eck et al.	N/A	N/A
9873316	12/2017	Lovold et al.	N/A	N/A
9944177	12/2017	Fischer	N/A	B60K 5/02
10023047	12/2017	Tanaka	N/A	N/A
10336409	12/2018	Girard et al.	N/A	N/A
D862286	12/2018	Flores	N/A	N/A
D866402	12/2018	Wang et al.	N/A	N/A
10569819	12/2019	Lovold et al.	N/A	N/A
D886678	12/2019	Lai et al.	N/A	N/A
D906181	12/2019	Lou et al.	N/A	N/A
D908536	12/2020	Ye et al.	N/A	N/A
D913847	12/2020	Hashimoto et al.	N/A	N/A
D935945	12/2020	Bruderer et al.	N/A	N/A
D950419	12/2021	Lou et al.	N/A	N/A
D950420	12/2021	Lou et al.	N/A	N/A
D958006	12/2021	Dunn et al.	N/A	N/A
D962823	12/2021	Dunn et al.	N/A	N/A
11718240	12/2022	Kelso	180/233	B60R 9/06
2002/0011745	12/2001	Petersen	N/A	N/A
2002/0047257	12/2001	Rondeau et al.	N/A	N/A
2002/0147072	12/2001	Goodell et al.	N/A	N/A
2002/0178968	12/2001	Christensen	N/A	N/A
2003/0001409	12/2002	Semple et al.	N/A	N/A
2004/0026150	12/2003	Nishi et al.	N/A	N/A
2004/0029459	12/2003	Berthiaume et al.	N/A	N/A
2004/0031640	12/2003	Tweet	N/A	N/A
2004/0079561	12/2003	Ozawa et al.	N/A	N/A
2004/0080186	12/2003	Crepeau et al.	N/A	N/A
2004/0108159	12/2003	Rondeau et al.	N/A	N/A
2004/0173653	12/2003	Audet	N/A	N/A
2004/0195797	12/2003	Nash et al.	N/A	N/A
2004/0239088	12/2003	Rondeau et al.	N/A	N/A
2005/0121953	12/2004	Sprouse	N/A	N/A
2005/0168018	12/2004	Cox	N/A	N/A
2005/0173177	12/2004	Smith et al.	N/A	N/A
2005/0173180	12/2004	Hypes et al.	N/A	N/A
2005/0247506	12/2004	Rondeau et al.	N/A	N/A
2005/0248116	12/2004	Fanson	N/A	N/A
2005/0275268	12/2004	Hiroyasu	N/A	N/A

2006/0066122	12/2005	Wiseman	N/A	N/A
2006/0097692	12/2005	Chen et al.	N/A	N/A
2006/0113139	12/2005	Nishi et al.	N/A	N/A
2006/0186699	12/2005	Davis et al.	N/A	N/A
2007/0034435	12/2006	Berg et al.	N/A	N/A
2007/0262604	12/2006	Takei et al.	N/A	N/A
2007/0278026	12/2006	Davis et al.	N/A	N/A
2008/0023240	12/2007	Sunsdahl et al.	N/A	N/A
2008/0023249	12/2007	Sunsdahl et al.	N/A	N/A
2008/0068849	12/2007	Ostroski	362/540	B60Q 1/24
2008/0217088	12/2007	Berg	N/A	N/A
2009/0195035	12/2008	Ripley et al.	N/A	N/A
2009/0236820	12/2008	Chang et al.	N/A	N/A
2010/0019524	12/2009	Utke et al.	N/A	N/A
2010/0084212	12/2009	Smith et al.	N/A	N/A
2011/0094813	12/2010	Suzuki et al.	N/A	N/A
2011/0094816	12/2010	Suzuki et al.	N/A	N/A
2012/0031694	12/2011	Deckard et al.	N/A	N/A
2014/0102820	12/2013	Deckard et al.	N/A	N/A
2014/0262584	12/2013	Lovold et al.	N/A	N/A
2015/0047917	12/2014	Burt et al.	N/A	N/A
2016/0332536	12/2015	Weber	N/A	B60G 15/062
2017/0023127	12/2016	Greasamar	N/A	E01H 5/061
2017/0066362	12/2016	Grider et al.	N/A	N/A
2017/0190373	12/2016	Quade et al.	N/A	N/A
2018/0009500	12/2017	Lovold	N/A	B62K 5/00
2018/0326843	12/2017	Danielson et al.	N/A	N/A
2018/0326847	12/2017	Landon	N/A	N/A
2019/0351836	12/2018	Chapdelaine et al.	N/A	N/A
2019/0367117	12/2018	Fischer	N/A	B60K 11/02
2020/0010120	12/2019	Kinsman et al.	N/A	N/A
2020/0171942	12/2019	Evans et al.	N/A	N/A
2020/0384932	12/2019	Cooper	N/A	B62J 15/00
2021/0188184	12/2020	Kelso et al.	N/A	N/A
2023/0331170	12/2022	Kelso et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3004855	12/2018	CA	N/A
317335	12/1955	CH	N/A
201076344	12/2007	CN	N/A
101934769	12/2010	CN	N/A
102015419	12/2010	CN	N/A
102802981	12/2011	CN	N/A
103101447	12/2012	CN	N/A
108463402	12/2017	CN	N/A
109415102	12/2018	CN	N/A
306390450	12/2020	CN	N/A
0116605	12/1899	DE	N/A
1755101	12/1970	DE	N/A

3033707	12/1981	DE	N/A
4136296	12/1992	DE	N/A
19531985	12/1996	DE	N/A
102005013610	12/2005	DE	N/A
0238077	12/1986	EP	N/A
0709247	12/1995	EP	N/A
0794096	12/1996	EP	N/A
1215107	12/2001	EP	N/A
1296036	12/2002	EP	N/A
1557345	12/2004	EP	N/A
1564123	12/2004	EP	N/A
0159650	12/1920	GB	N/A
1501631	12/1977	GB	N/A
06-105727	12/1993	JP	N/A
11-334447	12/1998	JP	N/A
2000-177434	12/1999	JP	N/A
2004-308453	12/2003	JP	N/A
03/53769	12/2002	WO	N/A
2008/100398	12/2007	WO	N/A
2008/115459	12/2007	WO	N/A

OTHER PUBLICATIONS

“2019 Polaris Sportsman 850” Polaris., posted date Jul. 23, 2019 [online], [retrieved on Jul. 13, 2021], Retrieved from the Internet <URL: <http://atvillustrated.com/content/2019-polaris-sportsman-850-sp-premium-review>> (Year: 2019). cited by applicant

“2020 Fourtrax Recon” Honda., posted date Jun. 22, 2019 [online], [retrieved on Jul. 13, 2021], Retrieved from the Internet <URL: <https://powersports.honda.com/atv/recutlity/fourtrax-recon?year=2020>> (Year: 2019). cited by applicant

“2022 Polaris Sportsman 450 H.O.” Polaris., posted date Jun. 3, 2021 [online], [retrieved on May 4, 2022], Retrieved from the Internet <URL: <https://premierpolaris.com/ATVs-Polaris-Sportsman-450-H-O-2022-Monroe-WA-e328a734-ecca-4aeb-ae9b-ad75007216abG>> (Year: 2021). cited by applicant

“2022 TRX250X” Honda., posted date Feb. 8, 2022 [online], [retrieved on May 4, 2022], Retrieved from the Internet <URL: <https://powersports.honda.com/atv/sport/trx250x>> (Year: 2022). cited by applicant

“LED Light Kit” Polaris., posted date Apr. 11, 2014 [online], [retrieved on May 4, 2022], Retrieved from the Internet <URL: https://atv.polaris.com/en-us/shop/accessories/lights/2884859/?utm_source=gmc&utm_medium=product_feed&utm_term=LED_Light_Kit> (Year: 2014). cited by applicant

“Sportsman 850” Polaris., posted date Jun. 29, 2017 [online], [retrieved on Jul. 13, 2021], Retrieved from the Internet <URL: <https://atv.polaris.com/en-us/sportsman-850/>> (Year: 2017). cited by applicant

International Preliminary Report on Patentability issued by the European Patent Office, dated Mar. 15, 2011, for International Patent Application No. PCT/US2009/059535; 6 pages. cited by applicant

International Preliminary Report on Patentability issued by the European Patent Office, dated May 11, 2009, for International Patent Application No. PCT/US2008/003483; 20 pages. cited by applicant

International Preliminary Report on Patentability issued by the European Patent Office, mailed Jan. 27, 2011, for International Patent Application No. PCT/US2009/049799; 13 pages. cited by

applicant

International Search Report and Written Opinion issued by the European Patent Office, dated Aug. 27, 2008, for International Patent Application No. PCT/US2008/003485; 11 pages. cited by applicant

International Search Report and Written Opinion issued by the European Patent Office, dated Jan. 6, 2010, for International Patent Application No. PCT/US2009/059535; 10 pages. cited by applicant

International Search Report and Written Opinion issued by the European Patent Office, dated Oct. 2, 2008; for International Patent Application No. PCT/US2008/003483; 15 pages. cited by applicant

International Search Report and Written Opinion issued by the European Patent Office, dated Sep. 1, 2008, for International Patent Application No. PCT/US2008/003480; 17 pages. cited by applicant

International Search Report and Written Opinion issued by the International Searching Authority, dated Sep. 19, 2016, for International Patent Application No. PCT/US2016/031992; 20 pages. cited by applicant

International Search Report issued by the European Patent Office, mailed Feb. 24, 2010, for International Patent Application No. PCT/US2009/049799; 6 pages. cited by applicant

The Council of the European Communities, “Council Directive on the approximation of the laws of the Member States relating to the driver's seat on wheeled agricultural or forestry tractors (78/764/EEC) (OJL 255, 18.9, 1978)”, Jul. 25, 1978, 69 pages, available at <http://eurlex.europa.eu/LexUriServ/LexUriServ.do?uri=CONSLEG:1978L0764:20070101:EN.pdf>. cited by applicant

Written Opinion issued by the International Searching Authority, mailed Feb. 24, 2010, for International Patent Application No. PCT/US2009/049799; 7 pages. cited by applicant

Primary Examiner: Phan; Hau V

Attorney, Agent or Firm: Faegre Drinker Biddle & Reath LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) The present application is a divisional of U.S. patent application Ser. No. 16/723,182, filed Dec. 20, 2019, the entire disclosure of which is incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to an all-terrain vehicle (hereinafter “ATV”) and, more particularly, to an ATV having improved ergonomics and performance.

BACKGROUND OF THE DISCLOSURE

(2) Generally, ATVs are used to carry one or two passengers and a small amount of cargo over a variety of terrains. Due to increasing recreational interest in ATVs, specialty ATVs, such as those used for trail riding, racing, and cargo hauling are in the market place. Most ATVs include an engine including between one and three cylinders. Most ATVs include a straddle or saddle type seat positioned above the engine. Examples of ATVs are disclosed in U.S. patent application Ser. No. 15/205,601, titled ALL-TERRAIN VEHICLE, filed Jul. 8, 2016, U.S. Pat. No. 9,873,316, titled ALL-TERRAIN VEHICLE, filed Mar. 13, 2014, U.S. Pat. No. 8,122,993, titled POWER STEERING FOR AN ALL TERRAIN VEHICLE, filed Aug. 14, 2008, and/or U.S. Pat. No. 8,215,694, titled ATV HAVING ARRANGEMENT FOR A PASSENGER, filed Oct. 7, 2008, the

entire disclosures of which are incorporated herein by reference.

SUMMARY OF THE DISCLOSURE

(3) In one exemplary embodiment of the present disclosure, an all-terrain vehicle is disclosed comprising a frame, a plurality of ground-engaging members supporting the frame relative to a ground surface, an outer body supported by the frame, and a cargo rack pivotably coupled to the outer body, the cargo rack including a hinge assembly comprising a first hinge leaf, a second hinge leaf complementary to the first hinge leaf, and a hinge pin inserted through the first hinge leaf and the second hinge leaf, wherein the cargo rack is moveable about the hinge pin between a closed position and an opened position.

(4) In another exemplary embodiment of the present disclosure, an all-terrain vehicle is disclosed comprising a frame, a plurality of ground-engaging members supporting the frame relative to a ground surface, a steering assembly operatively coupled to at least one of the plurality of ground-engaging members, the steering assembly configured to steer the at least one of the plurality of ground-engaging members, a battery supported by the frame, and a light pod supported by the steering assembly, the light pod including a battery tender port electrically coupled to the battery and configured to receive an electrical connector for providing power to the battery.

(5) In yet another exemplary embodiment of the present disclosure, an all-terrain vehicle is disclosed comprising a frame, a plurality of ground-engaging members supporting the frame relative to a ground surface, a seat assembly supported by the frame and configured to support an operator above the ground surface, a steering assembly adjacent the seat assembly and operatively coupled to at least one of the plurality of ground-engaging members, and a light pod supported by the steering assembly, the light pod including a side facing the steering assembly, and the light pod including at least one accessory port and an accessory plug inserted into the at least one accessory port.

(6) In yet another exemplary embodiment of the present disclosure, an all-terrain vehicle is disclosed comprising a powertrain having an air intake portion, a throttle body assembly in fluid communication with the air intake portion of the powertrain, the throttle body assembly including a throttle body and a cam rotatably coupled to the throttle body, and an adjustable throttle limiter operably coupled to the cam, the adjustable throttle limiter being moveable between a first position corresponding to a first angular displacement of the cam and a second position corresponding to a second angular displacement of the cam, and the first angular displacement greater than the second angular displacement of the cam.

(7) Additional features and advantages of the present invention will become apparent to those skilled in the art upon consideration of the following detailed description of the illustrative embodiments exemplifying the best mode of carrying out the invention as presently perceived.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above mentioned and other features of the invention, and the manner of attaining them, will become more apparent and the invention itself will be better understood by reference to the following description of embodiments of the invention taken in conjunction with the accompanying drawings.

(2) FIG. 1 is a left front perspective view of an illustrative ATV of the present disclosure;

(3) FIG. 2 is a left front perspective view of a front rack and storage area cover of the ATV of FIG. 1 in a fully closed position;

(4) FIG. 3 is a rear perspective view of the front rack and storage area cover of FIG. 2 in a partially opened position;

(5) FIG. 4 is an exploded view of a hinge assembly of the front rack and storage area cover of FIG.

3;

(6) FIG. 5 is a partial cross-sectional view of the hinge assembly of the front rack and storage area cover taken along lines 5-5 of FIG. 3 with the front rack and storage area cover in a fully opened position;

(7) FIG. 6 is a partial cross-sectional view of the hinge assembly of the front rack and storage area cover taken along lines 5-5 of FIG. 3 with the front rack and storage area cover in the fully closed position;

(8) FIG. 7 is a front left perspective of an electrical compartment of a front storage area of the ATV of FIG. 1;

(9) FIG. 8 is a top perspective view of the electrical compartment of FIG. 7;

(10) FIG. 9A is a partial exploded view of the electrical compartment of FIG. 7;

(11) FIG. 9B is a partial bottom perspective view of the electrical compartment of FIG. 7 mounted on a chassis of the ATV of FIG. 1;

(12) FIG. 10 is a front right perspective view of the electrical compartment of FIG. 7;

(13) FIG. 11 is a partial exploded view of a mounting bracket, an engine control unit, and a fuse pod or holder or fuse block of the electrical compartment of FIG. 10;

(14) FIG. 12 is a perspective view of the mounting bracket of FIG. 11;

(15) FIG. 13 is a left front perspective view of an illustrative throttle assembly with an adjustable throttle limiter of the present disclosure;

(16) FIG. 14A is a side view of the adjustable throttle limiter of FIG. 13 in a first position corresponding to a full throttle capacity of the throttle assembly of the present disclosure;

(17) FIG. 14B is a side view of the adjustable throttle limiter of FIG. 13 in a second position corresponding to a partial throttle capacity of the throttle assembly of the present disclosure;

(18) FIG. 14C is a side view of the adjustable throttle limiter of FIG. 13 in a third position corresponding to a minimum throttle capacity of the throttle assembly of the present disclosure;

(19) FIG. 15 is a perspective view of a plurality of multi-component routing clips of the ATV of FIG. 1;

(20) FIG. 16 is a perspective view of the routing clip of FIG. 15;

(21) FIG. 17 is a rear view of a light pod of the ATV of FIG. 1;

(22) FIG. 18 is a front perspective view of an accessory plug of the light pod of FIG. 17; and

(23) FIG. 19 is a rear perspective view of the accessory plug of FIG. 18.

(24) Corresponding reference characters indicate corresponding parts throughout the several views. Although the drawings represent embodiments of various features and components according to the present disclosure, the drawings are not necessarily to scale, and certain features may be exaggerated to better illustrate and explain the present disclosure. The exemplification set out herein illustrates an embodiment of the invention, and such an exemplification is not to be construed as limiting the scope of the invention in any manner.

DETAILED DESCRIPTION OF THE DRAWINGS

(25) The embodiments disclosed below are not intended to be exhaustive or to limit the invention to the precise forms disclosed in the following detailed description. Rather, the embodiments are chosen and described so that others skilled in the art may utilize their teachings. While the present disclosure is primarily directed to an ATV, it should be understood that the features disclosed herein may have application to other types of vehicles such as utility vehicles, motorcycles, watercraft, snowmobiles, people movers, and golf carts.

(26) With reference to FIG. 1, a vehicle 2 is shown as an all-terrain vehicle including a frame 4 coupled to and supported by front ground-engaging members 6 and rear ground-engaging members 8, ATV 2 travels relative to a ground surface on front ground-engaging members 6 and rear ground-engaging members 8. Front ground-engaging members 6 include wheels 6a and tires 6b, and rear ground-engaging members 8 include wheels 8a and tires 8b. ATV 2 generally includes a powertrain 10. Front ground-engaging members 6 are coupled to frame 4 by way of a front suspension 12, and

rear ground-engaging members **8** are coupled to frame **4** by way of a rear suspension **14**. ATV **2** further includes a seat assembly **16**, illustratively a saddle or straddle type seat **17**. While seat assembly **16** as shown in FIG. **1** is for a single rider or operator, it is contemplated that ATV **2** may be modified to incorporate two riders as described in U.S. Pat. Nos. 8,678,464 or 8,430,442, the disclosures of which are incorporated herein by reference. ATV **2** also includes a steering assembly **18** for steering at least front ground-engaging members **8**. Steering assembly **18** could be similar to that described in U.S. Pat. No. 8,122,993, titled POWER STEERING FOR AN ALL TERRAIN VEHICLE, filed Aug. 14, 2008. In the illustrative embodiment shown, steering assembly **18** includes handlebars **19**, illustratively left handlebar **19a** and right handlebar **19b**. Handlebars **19** could include at least one operator input for controlling one or more features or characteristics of ATV **2** such as, for example, the throttle.

(27) ATV **2** includes an outer body **20**, generally formed of a plastic material. Outer body **20** may include at least a footwell **22** for placement of a rider's foot while riding. ATV **2** also includes a forward storage area **24** positioned forward of steering assembly **18** and a rear storage area **26** positioned rearward of seat **17**. Rear storage area **26** includes a U-shaped or C-shaped rear cargo rack **28** configured to support cargo. In the embodiment shown, cargo rack **28** surrounds at least a portion of seat **17**. It should be appreciated that cargo rack **28** could include integrated tie downs such as those described in U.S. Pat. No. 8,905,435, the entire disclosure of which is incorporated herein by reference. As will be described in more detail herein, forward storage area **24** includes a front cargo rack **32** configured to support cargo. It should be appreciated that front rack **32** may also include a plurality of integrated tie downs. Outer body **20** may further include front body panel **33** including front fascia **34** and front fairing **36**, light pod **38**, inner panel **40**, side panel **42**, and fender body panels **44**. In various embodiments, front fascia **34** and front fairing **36** of front body panel **33** may be a single integral piece, while in other various embodiments, front fascia **34** and front fairing **36** of front body panel **33** may be multiple pieces.

(28) With reference now to FIGS. **2-12**, forward storage area **24** is shown in more detail. Referring initially to FIGS. **2** and **3**, front cargo rack **32** includes a generally flat upper surface **46** configured to support cargo. A plurality of handles **48**, illustratively a left handle **48a** and a right handle **48b**, are coupled to upper surface **46** and are configured to be grasped by an operator. Handles **48** may also comprise tie downs **49** for securing cargo to front rack **32**. In one embodiment, handles **48** are integrally formed with upper surface **46**; however, in other embodiments, handles **48** are removably coupled to upper surface **46**.

(29) Front rack **32** is pivotally coupled to front fascia **34** at a hinge assembly **50**. Accordingly, front rack **32** is pivotable between a fully closed position (see FIG. **2**) in which upper surface **46** of front rack **32** is generally parallel to a ground surface and a fully opened position (see FIG. **5**) in which front rack **32** is generally perpendicular to the ground surface. In this way, front rack **32** is generally pivotable between approximately 90 degrees. In various embodiments, front rack **32** may be configured to rotate beyond 90 degrees up to approximately 110 degrees such that front rack **32** is capable of remaining open on its own with the aid of gravity.

(30) Front rack **32** provides access to a forward storage bin or compartment **52**, which is sized and adapted to store cargo such as, for example, tools or gloves. In this way, front rack **32** is a lid or cover for storage compartment **52** and also forms part of outer body **20**. Front rack **32** is coupled to front body panel **33**, and includes at least one catch, illustratively a left catch **54a** and a right catch **54b**, for securing front rack **32** to front body panel **33**. Accordingly, front fascia **34** and/or front fairing **36** of front body panel **33** may include a matching latch (not shown) such as, for example, a flexible handle latch for engaging a respective one of catches **54a**, **54b**. It is contemplated, however, that front rack **32** may include one or more latches while front fascia **34** and/or front fairing **36** includes one or more corresponding catches. It is also contemplated that front rack **32** may securely latch to front fascia **34** and/or front fairing **36** using one or more locks. In embodiments, front rack **32** may latch to front fascia and/or front fairing **36** using a cable latch

system. Either of front rack **32** or a perimeter of storage compartment **52** may include a seal, or complementary parts of front rack **32** and storage compartment **52** may form a seal when front rack **32** is in the fully closed position, to assist with keeping water, dust, and other debris from entering storage compartment **52**. In this way, a seal is at least formed or defined when front rack **32** is in the closed position and adjacent or in contact with storage compartment **52**, front fascia **34**, and/or front fairing **36**.

(31) Referring now to FIGS. **4-6**, hinge assembly **50** of front rack **32** is shown in more detail. Hinge assembly **50** includes a hinge leaf **56** that is received within a complementary hinge leaf **58** of front fascia **34**. More specifically, a hinge pin **60** is inserted through hinge leaf **58** of front fascia **34** and hinge leaf **56** of front rack **32** to rotatably couple and secure front rack **32** to front fascia **34**. In this way, hinge pin **60** defines a hinge axis **62** of hinge assembly **50**, which is illustratively perpendicular to a longitudinal axis A (see FIG. **1**) of ATV **2** and is approximately perpendicular to vertical. In the embodiment shown, hinge pin **60** comprises an L-shaped body having a main or first leg **64** and a second leg **66** oriented generally perpendicular to first leg **64**. First leg **64** is arranged along hinge axis **62** and passes through hinge leaves **56**, **58**. A distal end of first leg **64** is adapted to facilitate passing first leg **64** through hinge leaves **56**, **58**. Second leg **66** defines both a handle and an axial stop for hinge pin **60** when first leg **64** is inserted into hinge leaves **56**, **58**. More particularly, when first leg **64** is inserted through hinge leaves **56**, **58**, second leg **66** abuts a portion of hinge leaf **56** and/or hinge leaf **58** to prevent further insertion of hinge pin **60**. While hinge pin **60** has been described as having a first leg and a second leg, it is contemplated that other designs of hinge pin **60** are possible to rotatably join front rack **32** and front fascia **34** and maintain the position of hinge pin **60** when front rack **32** is assembled with front fascia **34**.

(32) Hinge assembly **50** also includes at least one stop, illustratively a left stop **68a** and a right stop **68b**, configured to prevent front rack **32** from opening further than the fully opened position of FIG. **5**. Stops **68a**, **68b**, therefore, also indicate to a user that front rack **32** is in the fully opened position. In the embodiment shown, each of front rack **32** and front fascia **34** include complementary components of stops **68a**, **68b**. Referring specifically to FIGS. **5** and **6**, each of stops **68a**, **68b** includes a leg **70** extending distally from front rack **32**. Distally extending leg **70** includes a protrusion **72** proximate a distal end of leg **70**. Protrusion **72** is illustratively cylindrical and extends generally perpendicular to leg **70**. Front fascia **34** includes a channel or guide **76** sized and configured to receive at least one of leg **70** and protrusion **72**. In the embodiment shown, guide **76** is generally arcuate in shape and guides protrusion **72** as front rack **32** rotates about hinge axis **62** between the fully closed position and the fully opened position. More particularly, protrusion **72** and/or a portion of leg **70** rotates within guide **76** during movement of front rack **32**.

(33) Referring still to FIGS. **5** and **6**, front fascia **34** includes a surface **78** that protrusion **72** abuts when front rack **32** is in the fully opened position (see FIG. **5**). In the embodiment shown, surface **78** prevents front rack **32** from rotating about hinge axis **62** further than the fully opened position. In this way, surface **78** at least partially defines stops **68a**, **68b**. More particularly, the interaction of protrusion **72** with surface **78** may provide feedback to an operator that front rack **32** is in the fully opened position. In embodiments, guide **76** may include another surface **80** that protrusion **72** abuts when front rack **32** is in the fully closed position of FIG. **6**.

(34) Illustratively, at least one end **82** of guide **76** (e.g., an upper end) is open to facilitate disassembly of hinge assembly **50** and removal of front rack **32** from front fascia **34**. In the embodiment shown, hinge leaves **56**, **58** and stops **68** are integrally formed with the respective one of front rack **32** and front fascia **34** to form a single piece construction.

(35) Turning now to FIGS. **7-11**, an electrical compartment **84** of ATV **2** is shown in more detail. Electrical compartment **84** of ATV **2** is positioned within or adjacent front storage compartment **52**. Accordingly, electrical compartment **84** is positioned longitudinally forward of steering assembly **18** and houses, among other things, a battery **86**, an engine control unit ("ECU") **88**, a fuse compartment or holder **90**, and a mounting bracket **92**. In the embodiment shown, electrical

compartment **84** is accessible when front rack **32** is opened (e.g., the fully opened position of FIG. 5).

(36) Referring specifically to FIGS. **8-9B**, battery **86** is shown mounted within electrical compartment **84**. Battery **86** is received within a battery tray or well **94** and is secured using at least one battery hold down or clamp **96**. Battery well **94** illustratively comprises a molded plastic that surrounds or encapsulates battery **86**. As shown in FIG. **9A**, battery **86** is supported on a floor **95** of battery well **94**, which includes a plurality of corrugations **99** that provide a raised surface supporting battery **86** as well as a space for dirt, debris, or water to collect. Floor **95** also includes at least one drain hole **101** positioned intermediate corrugations **99** to provide an outlet for any water within battery well **94**.

(37) As shown in FIG. **9B**, battery well **94** is supported on a mounting bracket **103** of frame **4**. In some embodiments, battery well **94** may be coupled to mounting bracket **103** using, for example, one or more fastener. In other embodiments, battery well **94** may be supported by mounting bracket **103** while front fascia **34** is coupled to frame **4**. Mounting bracket **103** illustratively comprises a structural member of frame **4** and is coupled to rails **105** of frame **4**. In some embodiments, mounting bracket **103** may be coupled to rails **105** using at least one fastener. In other embodiments, mounting bracket **103** may be welded to rails **105**.

(38) In the embodiment shown, mounting bracket **103** comprises a box at least partially surrounding and supporting battery well **94**. A floor **107** of mounting bracket **103** includes at least one opening **109** sized to permit water from drain hole **101** to pass through. As a result, battery **86** is positioned longitudinally forward of a firewall **98** (see FIG. **7**) and steering assembly **18** of ATV **2**. Firewall **98** may be a panel or other assembly or component which separates at least the operator area (e.g., seat assembly **16**) from electrical compartment **84**. Illustratively, firewall **98** may be positioned adjacent a portion of steering assembly **18**, for example generally below a portion of handlebars **19**. An advantage, among others, of battery well **94** is a reduction in cost by incorporating battery well **94** into front fascia **34** while increasing the extent of protection provided to battery **96**.

(39) In the embodiment shown, battery **86** is intersected by longitudinal axis A of ATV **2** and is laterally oriented generally perpendicular to longitudinal axis A. It is contemplated, however, that battery **86** may be oriented parallel to longitudinal axis A or may be positioned substantially or entirely on one lateral side of longitudinal axis A.

(40) Battery clamp **96** comprises a laterally-extending body **100** having a flange **102** at a first end **104** configured to be received in a sidewall **106** of battery well **94** or electrical compartment **84** and a plurality of apertures **108** at a second end **110** of body **100**. Second end **110** of body **100** is secured to a flange **112** of battery well **94** or electrical compartment **84** with removable fasteners (not shown) to secure battery **86** within battery well **94**. In this way, battery **86** may be removed from vehicle **2** by pivoting cargo rack **32** to an open position, removing fasteners from second end **110**, and laterally moving clamp **96** to disengage flange **102** from first end **104** of sidewall **106**. With clamp **96** removed, battery **86** is accessible for repair, cleaning, replacing, testing, etc. Body **100** of clamp **96** illustratively comprises a rigid plastic such as, for example, a glass filled polypropylene. It is contemplated, however, that clamp **96** may comprise, for example, a rubber strap, bungee cord, or other retaining member. Battery clamp **96** also includes at least one vibration isolator **97** pressed into an upper surface of body **100**. Isolators **97** illustratively comprise a synthetic rubber, specifically ethylene propylene diene monomer ("EPDM") rubber configured to reduce the magnitude or extent of vibrations battery **86** is exposed to during operation of ATV **2**. Illustratively, battery **86** comprises a 12-volt valve-regulated lead-acid absorbent glass mat battery having protruding terminals, illustratively terminals **114a**, **114b**. It is contemplated, however, that battery **86** comprise a different type or shape of battery than illustrated.

(41) Referring now to FIGS. **10-12**, ECU **88**, fuse holder **90**, and mounting bracket **92** are shown in more detail. Illustratively, ECU **88**, fuse holder **90**, and mounting bracket **92** are positioned

longitudinally intermediate firewall **98** and battery **86**. Mounting bracket **92** is adjacent and/or mounted to firewall **98**, and ECU **88** and fuse holder **90** are supported by mounting bracket **92**. Accordingly, mounting bracket **92** includes an upper, generally vertical flange **116** and a lower, generally vertical flange **118**, which are configured to couple to firewall **98**. Specifically, flanges **116**, **118** each include at least one respective aperture **120**, **122** for receiving a fastener or stud for coupling to firewall **98**. A horizontal flange **124** extends outwardly from upper vertical flange **116** along longitudinal axis A of ATV **2** to join together upper and lower flanges **116**, **118**. In various embodiments, mounting bracket **92** may include an opening (not shown) configured to receive an accessory fuse relay, accessory fuse holder, or the like such that the accessory fuse relay, accessory fuse holder or the like may be attached to mounting bracket **92**.

(42) Lower flange **118** includes a vertical portion **126** which extends downwardly from horizontal flange **124** and a horizontal portion **128** which extends along longitudinal axis A toward firewall **98** from vertical portion **126**. Horizontal portion **128** includes at least one opening or aperture **144**, illustratively apertures **144a**, **144b**, **144c**, sized and shaped to route or guide a variety of components of ATV **2**. In the embodiment shown, apertures **144** of horizontal portion **128** route vent lines for a transmission of powertrain **10**, lines, wires, or conduits for a front drive of powertrain **10**, and conduits for a coolant bottle of a cooling assembly. Vertical flange **116** includes an engagement feature **146** for securing one or more electrical cables to mounting bracket **92** using, for example, a cable tie fastener. Vertical flange **116** also includes an opening **148** sized and shaped to permit components, illustratively a hose or conduit **111** for electrical wires for light pod **38**, of ATV **2** to pass therethrough. Firewall **98** may include an opening at least partially aligned with opening **148** of vertical flange **116** for the pass through of such components.

(43) As shown in FIG. **11**, fuse holder or fuse block **90** is mounted to horizontal flange **124** of mounting bracket **92** using at least one fastener **130**, illustratively a bolt. Fuse holder **90** includes a cover **140** removably coupled to a body **142**. Body **142** of fuse holder **90** may include one or more fuses that provide overcurrent protection of the electrical systems or subsystems of ATV **2**. Fuse holder **90** may also include one or more relays for the electrical system or subsystems of ATV **2**.

(44) ECU **88** is mounted to an aperture **134** of vertical portion **126** of mounting bracket **92** using at least one fastener **132**, illustratively a bolt. ECU **88** is electrically coupled to battery **86** and controls one or more electrical systems or subsystems of ATV **2**. ECU **88** may include one or more electrical ports **136**, **138** for electrically coupling to one or more electrical systems or subsystems of ATV **2** such as, for example, fuse holder **90**. An advantage, among others, of electrical compartment **84** is that battery **86**, ECU **88**, and fuse holder **90** may be grouped in close proximity to one another and may be accessed simultaneously when, for example, front rack **32** is in the fully opened position. Additionally, because such components are positioned in close proximity to each other, wires and lines extending between such components may be short. In addition, mounting bracket **92** permits at least two components, illustratively ECU **88** and fuse holder **90**, to be grouped together in close proximity to each other on the same mounting member.

(45) Turning now to FIGS. **13-14C**, a throttle assembly **150** of the present disclosure is illustrated. Throttle assembly **150** includes a throttle body **152** positioned intermediate an intake assembly **154** and an intake portion **156** of a power source, illustratively an engine. Throttle body **152** houses a throttle valve such as, for example, a butterfly valve rotatable about a throttle valve axis **158** to control an amount of intake air which enters intake portion **156** of the engine. More specifically, the throttle valve is rotatable about axis **158** in a direction **160** to increase the amount of air which enters intake portion **156** and is rotatable about axis **158** in a direction **162** to decrease the amount of air which enters intake portion **156**. The rotation of the throttle valve about axis **158** may be controlled by a biasing member (not shown) which biases throttle valve in direction **162** and by an input on handlebars **19** (e.g., a throttle lever) via a throttle cable **164**.

(46) Throttle cable **164** is received within a circumferential groove **168** of a cam or pulley **166** fixedly coupled to the throttle valve. In this way, throttle cable **164** at least partially encircles pulley

166, which is rotatable about axis **158** in directions **160**, **162** with the throttle valve. An end of throttle cable **164** includes a crimp or weldnut **165** that is received within an opening or slot **167** of pulley **166**. Throttle cable **164** is axially displaceable along an axis **170** tangent to circumferential groove **168** of pulley **166** in directions **172**, **174**. More specifically, when throttle cable **164** is displaced in direction **174** as a result of, for example, an operator input to handlebars **19**, pulley **166** is rotated about axis **158** in direction **160** to increase the amount of air provided to intake portion **156** of the engine. When the biasing member biases the throttle valve and pulley **166** in direction **162**, throttle cable **164** is displaced in direction **172**.

(47) Throttle assembly **150** also includes an adjustable throttle limiter **176** coupled to pulley **166** and configured to control or limit an extent of rotation of the throttle valve and pulley **166** about axis **158** in direction **160**. More specifically, throttle limiter **176** is adjustable between a first amount of rotation of the throttle valve corresponding to full throttle access and a second amount of rotation of the throttle valve corresponding to no throttle access. In embodiments, the second amount of rotation of the throttle valve may be no rotation of the throttle valve about axis **158**.

(48) Throttle limiter **176** comprises a cylindrically shaped housing **178**. An exterior surface of housing **178** includes a plurality of engagement features **180**, illustratively threads. In the embodiment shown, threads **180** of housing **178** engage with a weldnut **182** of a mounting bracket **184** to align housing **178** with pulley **166** and permit housing **178** to be axially adjusted relative to mounting bracket **184**. A fastener **188**, illustratively a lock nut, engages threads **180** of housing **178** opposite bracket **184** and weldnut **182**. When lock nut **188** is rotated to engage mounting bracket **184**, housing **178** is prohibited from axially displacing toward mounting bracket **184**. Mounting bracket **184** is coupled to throttle body **152** and provides a reference point or datum **186** (see FIGS. **14A-14C**) for adjusting housing **178** along an axis **190** in directions **192**, **194** relative to mounting bracket **184** and pulley **166**. Throttle limiter **176** also includes a cable **196** that passes through housing **178** and is received within circumferential groove **168** of pulley **166**. A first end of cable **196** includes a crimp or weldnut **198** that is received within an opening or slot **169** of pulley **166**. As a result, when pulley **166** rotates about axis **158** in direction **160**, cable **196** is axially displaced in direction **194**. Conversely, when pulley **166** rotates about axis **158** in direction **162**, cable **196** is axially displaced in direction **192**.

(49) A second end of cable **196** includes a crimp or weldnut **200** secured by a biasing assembly **202**. Biasing assembly **202** is positioned intermediate weldnut **200** of cable **196** and an adjustment end **204** of housing **178**. Biasing assembly **202** includes a biasing member **206**, illustratively a compression spring, positioned intermediate opposing end caps **208**, illustratively end caps **208a**, **208b**. Biasing member **206** applies a biasing force tending to spread ends caps **208a**, **208b** apart from one another. Because end cap **208b** abuts adjustment end **204** of housing **178**, biasing member **206** biases end cap **208a** in direction **192** away from end cap **208b**, indicated by a variable distance **210** in FIGS. **14A-14C**. In the embodiment shown, distance **210** generally corresponds to an extent of angular displacement of pulley **166** about axis **158** in direction **160**. Thus, the larger distance **210** is, the larger the angular displacement of pulley **166** and the more air that enters intake portion **156** of the engine. Conversely, the smaller distance **210** is, the smaller the angular displacement of pulley **166** and the more air that enters intake portion **156** of the engine. In this way, throttle limiter **176** controls and modulates the throttle body **152** and the performance characteristics of the vehicle.

(50) Because weldnut **198** is secured to pulley **166** and pulley **166** is a fixed distance from mounting bracket **184**, a distance **212** of adjustment end **204** of housing **178** relative to mounting bracket **184** (and datum **186**) influences an extent of distance **210**. In the embodiment shown, distance **212** and distance **210** have an inverse relationship. For example, the smaller distance **212** is, the larger distance **210** is (see FIG. **14A**); and the greater distance **212** is, the smaller distance **210** is (see FIG. **14C**). Adjustment end **204** of housing **178** includes an engagement feature **214** configured to be grasped by an operator or a tool to rotate housing **178** about axis **190**. Rotation of

threaded housing **178** about axis **190** results in axial displacement of housing **178** along axis **190** in directions **192**, **194** depending upon the direction of rotation of housing **178** about axis **190**. When housing **178** is rotated about axis **190** to displace housing **178** along axis **190** in direction **192**, distance **212** increases and distance **210** decreases thereby reducing the amount of air which can enter intake portion **156** of the engine. When housing **178** is rotated about axis **190** to displace housing **178** along axis **190** in direction **194**, distance **212** decreases and distance **210** increases thereby increasing the amount of air which enter intake portion **156** of the engine. Accordingly, FIG. **14A** illustrates a position of throttle limiter **176** corresponding to the maximum amount of air which can enter intake portion **156** of the engine, FIG. **14B** illustrates a position of throttle limiter **176** corresponding to a partial amount of air which can enter intake portion **156** of the engine relative to the position of FIG. **14A**, and FIG. **14C** illustrates a position of throttle limiter **176** corresponding to a minimum amount of air which can enter intake portion **156** of the engine.

(51) In the embodiment shown, both throttle cable **164** and cable **196** of throttle limiter **176** are mounted to mounting bracket **184**. It is contemplated, however, that only cable **196** of throttle limiter **176** may be mounted to mounting bracket **184**. Throttle limiter **176** may be located internally of the outer body of the vehicle. In one embodiment, throttle limiter **176** is accessible through the front, left wheel well. In addition, a sensor may relay information regarding the position of throttle limiter **176** to an operator or a display screen of light pod **38**.

(52) Turning now to FIGS. **15** and **16**, ATV **2** includes a plurality of multi-component routing clips **216** coupled to a frame rail **218** of frame **4**. Each of routing clips **216** are configured to secure one or more components of ATV **2** along frame **4**. More specifically, routing clip **216** includes an elongated body **220** having a mounting flange **222** for coupling to frame **4**. In the embodiment shown, mounting flange **222** has a vertical portion **222a** and a horizontal portion **222b**. Vertical portion **222a** includes an aperture **224** sized to receive a fastener, for example, a push dart style fastener, for coupling routing clip **216** to frame **4**. Similarly, horizontal portion **222b** may include an aperture sized to receive a fastener, for example, a push dart style fastener, for coupling routing clip **216** to frame **4**. In this way, routing clip **216** may be coupled to frame **4** with at least one of vertical portion **222a** and horizontal portion **222b** of mounting flange **222**.

(53) Body **220** includes an elongated portion **226** extending from horizontal portion **222b** of mounting flange **222**. Elongated portion **226** includes a planar portion **228**, a plurality of snaps or clips **230**, illustratively clips **230a-d**, and a cradle **232** at a distal end of elongated portion **226**. Planar portion **228** is positioned intermediate horizontal portion **222b** of mounting flange **222** and intermediate clips **230a-d**. Planar portion **228** includes at least one slot **234**, illustratively slots **234a**, **234b**, configured to receive a fastener such as, for example, a cable tie. In this way, at least one electrical cable or wiring harness **233** of ATV **2** may be secured to planar portion **228** of routing clip. Clips **230a-d** are positioned intermediate planar portion **228** and cradle **232** and are sized and shaped to receive and secure, for example, an electrical cable **235** or an Evaporative Emission Control (hereinafter “EVAP”) System hose **237**. Each of clips **230a-d** is circular in shape and includes a respective springed opening **236a-d**. Springed openings **236a-d** include flexible protrusions **238**, which provide a positive indication for an assembler that a respective component of ATV **2** is aligned with the respective one of springed openings **236a-d**. Flexible protrusions **238** also provide a biasing force to retain the component of ATV **2** within the respective one of clips **230a-d**. In the embodiment shown, clips **230a-d** are sized to secure a corresponding component of ATV **2**. Accordingly, a diameter of clip **230a** is smaller than a diameter of clips **230b-d**. It is contemplated, however, that clips **230a-d** may be the same size or a different size than shown in FIGS. **15** and **16**. It is also contemplated that routing clip **216** may include more or fewer than clips **230a-d**.

(54) Arcuately shaped cradle **232** (e.g., generally a “C” or “U” shape) is sized and adapted to conform to an outer circumference of a coolant hose **239**. Cradle **232** includes at least one slot **240** configured to receive a fastener such as, for example, a cable tie for securing coolant hose **239** (or

another component of ATV 2) to cradle 232. In other embodiments, cradle 232 may include springed openings and flexible protrusions similar to clips 230a-d. In this way, at least cradle 232 is configured to accommodate conduits, lines, wires, and/or tubes of different diameters because the fastener (e.g., a cable tie) can be lengthened or shortened to secure such members within cradle 232. Elongated body 220 of routing clip 216 also includes a laterally extending side tab 242 positioned laterally adjacent clips 230a-d. In the embodiment shown, side tab 242 provides a mounting location for a component of ATV 2 such as, for example, an air breather for the engine. Routing clip 216 may comprise a polymer injected molded part.

(55) Turning now to FIGS. 17-19, light pod 38 is shown in more detail. A side of light pod 38 facing seat assembly 16 includes an information display 244 having a display screen 246 configured to display information regarding ATV 2 to an operator. Information display 244 may also include at least one operator input 248, such as a push button, to interact with the information displayed on information display 246. Display screen 246 may be a touch-screen display configured for two-way input such that the operator may input selections through display screen 246 and received information from display screen 246. Alternatively, display screen 246 may merely be a one-way screen configured to display information only.

(56) Positioned adjacent information display 244 is at least one accessory port 250, illustratively accessory ports 250a-d. Accessory ports 250a, 250b include an operator input 251, illustratively a switch, configured to operate an accessory of ATV 2, and accessory ports 250c, 250d include an accessory plug 252. Accessory plug 252 is inserted into and covers unused accessory ports 250 but may be removed to add a switch when an accessory is added to ATV 2. Accessories include, but are not limited to, a winch and auxiliary lighting. An advantage, among others, of accessory plug 252 is that unused accessory port 250 may be covered until an accessory switch needs to be added to light pod 38. The accessory switch can be added to light pod 38 without modifying light pod 38. As shown in FIGS. 18 and 19, accessory plug 252 comprises a generally planar cover 254. Extending rearward from planar cover 254 are at least one retaining member 256, illustratively retaining members 256a, 256b, for retaining accessory plug 252 in accessory port 250. Accessory plug 252 also includes a tab 258 extending laterally outward from planar cover 254. Tab 258 facilitates removal of accessory plug 252 from accessory port 250.

(57) Referring specifically to FIG. 17, light pod 38 also includes an ignition switch 260 and a battery tender port 262. Battery tender port 262 is electrically coupled to battery 86 and permits an operator to charge battery 86 when ATV 2 is not in use. That is, battery tender port 262 is configured to receive a complementary electrical connector from a battery tender and transfer power from the battery tender to battery 86. In the embodiment shown, battery tender port 262 is positioned vertically below display screen 246 and horizontally aligned with ignition switch 260. An advantage, among others, of battery tender port 262 positioned on light pod 38 is that battery tender port 262 is in an easily accessible location for charging battery 86. Another advantage, among others, of battery tender port 262 positioned on light pod 38 is that battery tender port 262 is readily visible to an operator and provides a visual reminder to charge battery 86. Battery tender port 262 may include a cover 264 configured to protect battery tender port 262 when not in use such as, for example, during operation of ATV 2.

(58) Light pod 38 further includes a 12v accessory socket or port 266 and an off switch 268. Accessory port 266 is configured to receive a complementary adapter or plug and provide electrical power from the electrical system of ATV 2 to a portable accessory. Accessory port 266 may include a cover 270 configured to protect battery tender port 262 when not in use. Off switch 268 is controllably coupled to, for example, powertrain 10 or battery 86 and includes a base 272 and a leash or tether 274 releasably coupled thereto. When tether 274 is detached from base 272 during operation of ATV 2, powertrain 10 may be shut off to prohibit ATV 2 from further operation until tether 274 is reattached to base 272. In some embodiments, ATV 2 may not be operable until and unless tether 274 is attached to base 272.

(59) While this invention has been described as having an illustrative design, the present invention may be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains.

Claims

1. An all-terrain vehicle, comprising: a frame; a plurality of ground-engaging members supporting the frame relative to a ground surface; a steering assembly operatively coupled to at least one of the plurality of ground-engaging members, the steering assembly configured to steer the at least one of the plurality of ground-engaging members; a battery supported by the frame; and a light pod supported by the steering assembly, the light pod including a battery tender port electrically coupled to the battery and configured to receive an electrical connector for providing power to the battery.
2. The all-terrain vehicle of claim 1, further comprising a seat assembly adjacent the steering assembly, and the battery tender port is positioned on a side of the light pod facing the seat assembly.
3. The all-terrain vehicle of claim 1, wherein the battery tender port includes a removable cover.
4. The all-terrain vehicle of claim 1, further comprising a display screen and an ignition switch, and the ignition switch and the battery tender port are positioned vertically below the display screen.
5. The all-terrain vehicle of claim 4, wherein the display screen is operatively coupled to at least one operator input, and the at least one operator input is configured to allow an operator to interact with information displayed on the display screen.
6. The all-terrain vehicle of claim 4, wherein the display screen is a touch screen display.
7. The all-terrain vehicle of claim 1, further including a seat assembly supported by the frame.
8. The all-terrain vehicle of claim 7, wherein the light pod includes a first side facing toward the seat assembly and a second side opposite the first side, and the battery tender port is positioned on the first side.
9. The all-terrain vehicle of claim 1, wherein the light pod further includes at least one accessory port configured to receive an accessory plug of an accessory.
10. The all-terrain vehicle of claim 9, wherein the accessory includes at least one of a winch and an auxiliary light.
11. An all-terrain vehicle, comprising: a frame; a plurality of ground-engaging members supporting the frame relative to a ground surface; a seat assembly supported by the frame and configured to support an operator above the ground surface; a steering assembly adjacent the seat assembly and operatively coupled to at least one of the plurality of ground-engaging members; and a light pod supported by the steering assembly, the light pod including a first side facing the steering assembly, and the light pod including at least one accessory port positioned on the first side of the light pod and an accessory plug configured to be inserted into the at least one accessory port.
12. The all-terrain vehicle of claim 11, wherein the accessory plug includes at least one retaining member for retaining the accessory plug in the at least one accessory port.
13. The all-terrain vehicle of claim 11, wherein the first side of the light pod includes an information display, and the at least one accessory port is positioned adjacent the information display.
14. The all-terrain vehicle of claim 13, wherein the display screen is operatively coupled to at least one operator input, and the at least one operator input is configured to allow an operator to interact with information displayed on the display screen.
15. The all-terrain vehicle of claim 13, wherein the display screen is a touch screen display.
16. The all-terrain vehicle of claim 11, wherein the accessory plug is operatively coupled to an

accessory, and the accessory includes at least one of a winch and an auxiliary light.

17. An all-terrain vehicle, comprising: a frame; a plurality of ground-engaging members supporting the frame; a seat assembly supported by the frame; a steering assembly operatively coupled to at least one of the plurality of ground-engaging members; a battery supported by the frame; and a light pod supported by the steering assembly, the light pod including a battery tender port operatively coupled to the battery and configured to receive a battery tender, the battery tender configured to transfer power to the battery.

18. The all-terrain vehicle of claim 17, further including a display screen supported by the light pod and facing the seat assembly.

19. The all-terrain vehicle of claim 17, wherein the battery tender port is positioned on a side of the light pod facing the seat assembly.

20. The all-terrain vehicle of claim 17, wherein the battery is configured to be charged by the battery tender when the all-terrain vehicle is not in use.
